

exhibit these qualities.

The advantages in speed and high-frequency performance of GaAs are slightly offset by the manufacturing process limitations of GaAs, in terms of both complexity and reproducibility as opposed to silicon. Moreover, once a company has set up its fabrication and manufacturing process for silicon, they are extremely unlikely to undertake a major and risky change to GaAs because fabrication processes of GaAs are large and complicated.

Recent initiatives

GaAs technology has been accepted as vital and strategic to the future development of the economy and world economies have promised to make the technology viable in near future. GaAs worldwide demand at \$3.8 billion in 2020 is expected to grow to \$22 billion by 2026. GaAs components will achieve broad economies of scale, thus making them more affordable and available.

Less power consumption and more efficiency offered by GaAs wafers are attracting the market players to adopt these wafers, thereby increasing the demand for GaAs wafers. Previously, the optoelectronic devices were broadly used in short-range optical communications and computer peripherals but now these are in demand for some emerging applications such as lidar, augmented reality, and face recognition.

The initiatives taken by different countries, companies, and institutes towards the development of GaAs based technology are briefly summarised below.

- Many countries like Taiwan, Korea, Singapore, and China, who are the major importers of chemical elements and compounds doped for use in electronics (in the form of discs, wafers, cylinders, and rods), occupied a share of 56.1% in the world imports as of 2018. These countries had captured the leading share of 42.84% in the global GaAs market during FY2018. Increased focus of Chinese companies in manufacturing 10cm and 15cm (4- and 6-inch) GaAs based vertical-cavity surface-emitting laser-epitaxial layered (VCSEL EPI) wafers is boosting the use of these wafers in both 3D and data communi-

cation applications. Taiwan players, particularly dealing with the supply of GaAs wafers and manufacturers of vertical-cavity surface emitting laser components, are collectively growing the GaAs market value in the region.

- Mitsubishi Chemical, DOW Electronics, Sumitomo Electric, AXT, Freiberg Compound Materials, Advanced Wireless Semiconductor, IQE, Win Semiconductors, Global Communication Semiconductors, and Intelligent Epitaxy Technology are some of the primary players that are working in the global GaAs wafer market.

- WIN Semiconductors Corp. (Taiwan) has become the first pure-play 15cm GaAs foundry in the world and has completed its phase 2 production capacity expansion of newest wafer fabrication. The increased GaAs wafer manufacturing capacity by 20% has been possible with the usage of efficient process lines and advanced equipment in GaAs MMIC production. This company is well-known for producing GaAs microwave integrated circuits on 15cm wafers with the help of advanced manufacturing technology and production equipment.

- EVG (Austria) is also a noticeable supplier of wafer bonding and lithography equipment, and the company bears a good brand value in the GaAs wafer market. EVG announced its partnership with Ningbo Semiconductor International Corporation (NSI) in China, and this business strategy is targeted to develop the industry's first process technology platform that deals with wafer-level heterogeneous integration of GaAs on silicon to be useful in the RF front-end module (FEM) making. This is ultimately helpful in the development of next-generation, high-performance, and ultra-compact RF front-end chipsets that are highly in demand for 4G and 5G smartphones and other handsets.

- G-ray Nanotech and Das Leibniz-Institut für Kristallzüchtung (IKZ) in Berlin are gradually gaining prominence in the GaAs wafer market. In February 2019, G-ray Nanotech and IKZ stepped into a research and development collaboration with more focus towards the doping of GaAs structures and in the manufacturing of high-purity crystals in wafer form. Such crystals

are useful in detector applications.

- A British semiconductor company (IQE) has announced that the capability of 15cm GaAs diameter wafer is enabling its customers and also helping in reducing the unit cost of VCSEL chips, thereby accelerating the adoption of this technology.

Indian initiatives

India is emerging as the next major semiconductor chip designer and manufacturer in the world and, according to the Indian Electronics & Semiconductor Association, the Indian semiconductor industry, growing at 10% annually, is set to be worth \$33 billion by 2025. India is building an electronics system design manufacturing industry with more than 30,000 engineers working on different aspects of chip design and verification, and currently nearly 3,000 chips are designed every year.

The government of India is taking initiatives to boost the economy through schemes such as Make in India, 100% foreign direct investment (FDI), and support of subsidies and incentives. Almost all defence laboratories, institutes, and universities of national repute are working in the field of GaAs semiconductor applications. Some of their initiatives are summarised below.

- Hyderabad based DRDO facility working on GaAs devices' fabrication is important from the defence R&D point of view.

- Research initiative on non-silicon based technologies (such as GaAs) for nanofabrication and nanoscale devices has been taken up at IIT Delhi. The project involves setting up of a nanofabrication facility for the development of nanofabrication processes and their use towards making non-silicon nanoscale devices.

- Centre for nanoelectromechanical systems (NEMS) and nanophotonics at IIT Madras has been set-up and is pursuing R&D to achieve nanophotonics components/systems based on GaAs/AlGaAs. **EFY**



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